

JOHNATHON SOUTHWORTH

Senior Platform Engineer | Full-Stack & Systems Architecture

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Full-stack engineer with 8 years of experience architecting and maintaining a CRM-integrated business platform spanning frontend, backend, integrations, security, and AI.

EXPERIENCE / SKILLS

Playcraft Systems

May 2021 - Present

Senior Platform Engineer

Sole technical owner of the dealer platform, meeting monthly with sales consultants to turn field pain points into shipped features for 12 distribution groups and 300+ external users.

Applied AI & Automation | Performance Engineering | Identity & Access

- Designed and shipped a HubSpot-connected dealer operations platform that avoided 60+ Sales Hub seats and over \$75K in recurring annual licensing.
- Built and deployed a production AI voice/text agent using VAPI with retrieval-augmented generation (RAG) and human-in-the-loop escalation, fully automating order-status lookups and eliminating 25+ weekly service emails for installation-instruction and product-drawing requests.
- Developed the playground industry's first dynamic 3D configurator utilizing Babylon.js, engineering a custom component-compatibility rules engine and real-time ASTM/CSA safety compliance overlays that empowered dealers to generate instant sales collateral.
- Resolved a critical N+1 query cascade in a production catalog endpoint, slashing database queries from ~337 down to 8 and reducing response payloads from an unmanageable ~99 MB to under 1 KB, successfully restoring sub-second performance.
- Architected multi-tier Role-Based Access Control (RBAC), token-based authentication, and strict tenant isolation to securely enforce complex dealer data-visibility rules across a shared multi-tenant platform.
- Replaced legacy SharePoint silos with a centralized, AWS-backed DAM platform processing 70K+ source assets and 400K+ derivatives, with automated image-transformation pipelines and semantic search that enabled instant cross-departmental asset discovery and eliminated manual image scaling for the graphics & marketing teams.
- Architected an OCR extraction pipeline that parses complex fabrication blueprints, algorithmically isolating cut and bend specifications to auto-generate job processor reports and eliminating dozens of hours of weekly manual drawing review for the fabrication department.

Playcraft Systems

May 2018 - May 2021

Software Engineer

Partnered with leadership and department heads across sales, production, and marketing to build workflows that reduced assembly time for installation documents.

Workflow Automation | Business Intelligence | Web Architecture

- Engineered a BOM-driven document generation pipeline that cut weekly assembly time from ~30 hours to 2 hours, accelerating the sales cycle through instant, self-serve document retrieval.
- Built a real-time sales analytics dashboard with React.js, a RESTful API, and D3.js, giving leadership the distributor insights to forecast revenue and identify new market opportunities.
- Developed a shop-floor time-tracking application that digitized engineering workflows and established performance KPIs based on user skill and part complexity, improving production efficiency.
- Owned the architecture of internal and public platforms end to end, covering database administration, CDN configuration, and backend security while maintaining high availability and accessibility compliance.

TECHNICAL SKILLS

Languages & Frameworks

TypeScript, PHP (Yii2), Node.js, React, Next.js, Babylon.js, D3.js

Data, Cloud & APIs

MS SQL Server, MySQL, Firebase/Firestore, GCP, AWS, HubSpot, VAPI, Stripe, Twilio

EDUCATION & CERTIFICATIONS

Rogue Community College

June 2018

AAS, Web Development

CompTIA

August 2024

Security+ Certification